

Aakash Shahani

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EDUCATION

University of South Florida

Bachelor of Science in Computer Science

Tampa, FL

Graduated: Dec 2025

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Distributed Systems, Database Design, Software Testing, Object-Oriented Programming.

EXPERIENCE

Research Assistant (Computational Systems)

CSSAI Lab, University of South Florida

Tampa, FL

Oct 2025 – Present

- Architected a real-time framework for **Human-AI Teaming** using Python and **WebSockets**, reducing synchronization latency to **50ms** between human inputs and AI agents.
- Developed scalable data pipelines to capture and serialize interaction logs from **1M+ nodes**, enabling real-time performance bottleneck analysis and system monitoring.
- Refactored simulation modules for complex negotiation environments, improving execution runtime by **25%** through efficient state management and multi-threading.

AI Research Assistant (Alzheimer's Disease Prediction)

University of South Florida

Tampa, FL

Aug 2024 – Apr 2025

- Optimized Deep Learning models for identifying protein kinase drug targets in **Alzheimer's patients**, achieving a **20% increase** in predictive accuracy via hyperparameter tuning.
- Engineered automated **ETL scripts** to clean and annotate multi-million record biological datasets, improving data quality metrics by **30%** for model training pipelines.

PROJECTS

Distributed URL Shortener | Python, Redis, AWS, Docker

Dec 2025

- Designed a high-throughput **RESTful** service with **Redis caching**, reducing database read load by **80%** and ensuring **sub-100ms** request latency.
- Architected for horizontal scalability using **consistent hashing** to support high concurrency across microservices.

Real-Time Collaborative Whiteboard | Go, WebSockets, React, Redis

Nov 2025

- Built a low-latency collaboration tool using WebSockets and implemented **CRDTs** to resolve concurrent edit conflicts with **zero data loss**.
- Utilized Redis AOF for persistent state recovery, ensuring high availability during server failover events.

Stock Market Analysis Engine | Python, Yahoo Finance API, Plotly

May 2023

- Engineered a data ingestion engine using the **Yahoo Finance API** to fetch and process daily OHLCV data, implementing error handling for API rate limits and data gaps.
- Developed **multi-threaded pattern recognition algorithms** to detect 12+ candlestick formations with **89% accuracy** via backtesting on **7 years** of historical data.

Scalable Task Scheduler | Python, Multithreading, PostgreSQL

Aug 2025

- Designed a custom scheduling system supporting priority queues and retries using Python's **multiprocessing**, increasing throughput for high-volume workloads.
- Optimized relational schemas to track task execution states, maintaining **99.9% data integrity** under heavy load.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (PostgreSQL, MySQL), Go, HTML/CSS

Tools & Frameworks: **REST**, AWS, Docker, Redis, WebSockets, CI/CD, React, FastAPI, Flask, PyTest, Git, Linux, Kubernetes

Core Competencies: RESTful API Design, Microservices, Asynchronous Programming, Test-Driven Development (TDD), Documentation (README/Swagger), Version Control (Gitflow)